



Robert Hendrich, Lukas Pilz und Leon Frey

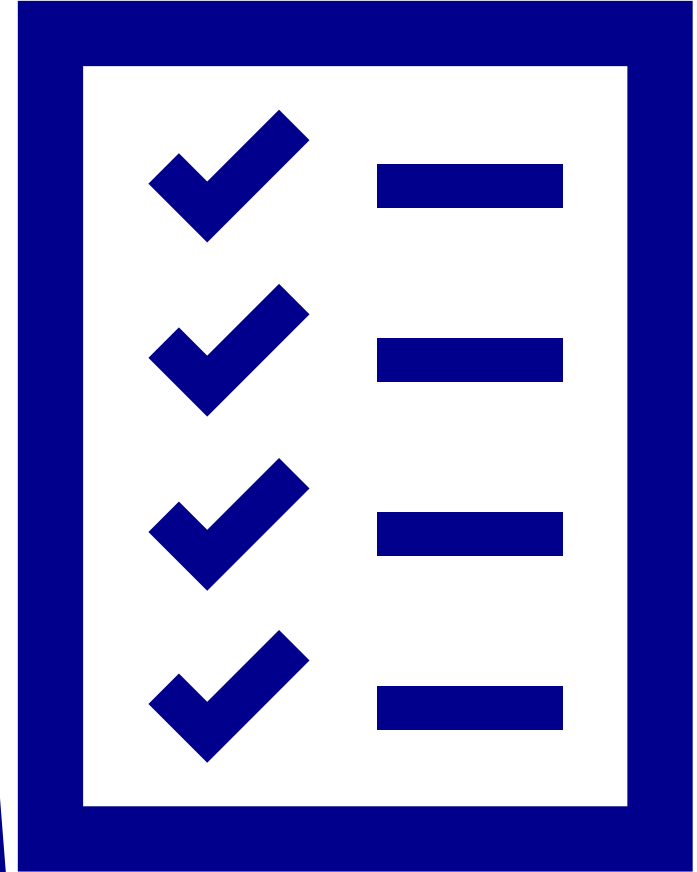
Dresden, 19. Mai 2026

Gestaltungsansätze der Advanced Data Analytics

Group 3: Cycling Stage Prediction

Gliederung

1. Aufgabenstellung (Leon)
2. Datenbeschaffung
 - 2.1 Proccyclingstats (Lukas)
 - 2.2 Geo- und Wetterdaten (Leon)
3. Datenexploration (Robert)
4. Ausblick (Lukas)



1. Aufgabenstellung

Topic 1: Cycling stage prediction

Ziel: Vorhersage der Top-10/Top-20-Platzierten je Etappe (Fahrerranking nach Abschlussplatzierung)

- 1) **Data collection, integration, and cleaning** (e.g., filtering, harmonization, noise removal, removal of duplicates, missing values, feature scaling, etc.)
- 2) **Descriptive data analysis for data exploration & data understanding** (e.g., dataset properties, distribution, univariate, bivariate, multi-variate, correlations, imbalance, etc.)
- 3) **Development of an automated machine learning pipeline** (feature preparation, sampling/data splitting, model training, evaluation strategy, etc.)
- 4) **Implementation of different machine learning models**
 - **Simple models** (e.g., *linear models, decision tree*)
 - **Advanced flexible models** (e.g., *tree-based ensembles, deep neural networks, tabular foundation models*)
 - **Advanced interpretable models** (e.g., *generalized additive models*)
- 5) **Model evaluation, diagnosis, interpretation** (post-hoc explanations, interpretable model structure)





2. Datenbeschaffung

2.1 Proccyclingstats

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Scraping Rennergebnisse via <https://www.procyclingstats.com/>

1. Etappeergebnisse der Grand Tours 2005-2025

- Das Datenvolumen: 3 Rennen * 21 Jahre * 21 Etappen = ca. 1.323 Einzeletappen
- Nutzung von:
 - Curl_cffi als HTTP client
 - Selectolax parser
- Output:
 - JSONL File



```
{
  "race": "tour-de-france",
  "year": 2024,
  "url": "https://www.procyclingstats.com/race/tour-de-france/2024/stage-1",
  "metadata": {
    "departure": "Florence",
    "arrival": "Rimini",
    "distance": "206 km",
    "vertical_meters": "3847",
    "profile_score": "4",
    "won_how": "Solo",
    "avg_speed": "41.51 km/h"
  },
  "results": [
    { "rank": "1", "rider_url": "romain-bardet", "time_gap": "5:07:22" },
    { "rank": "2", "rider_url": "frank-van-den-broek", "time_gap": ",," }
  ]
}
```

2. Fahrerprofile

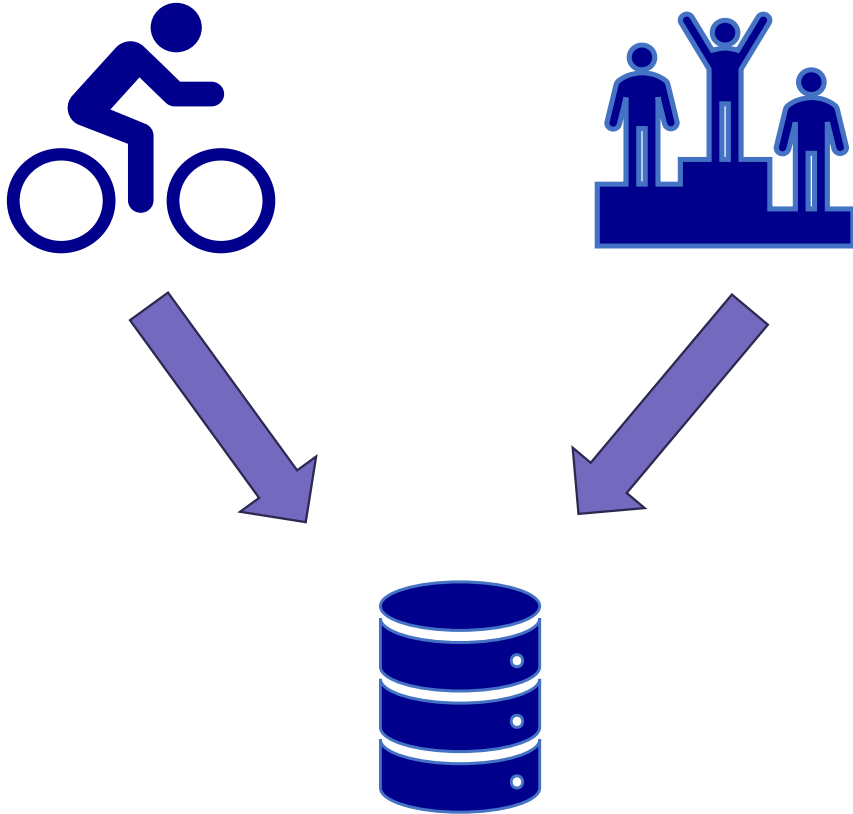
- Basis = Ergebnisdaten
 - Extraktion Fahrer URL, wenn in den Ergebnislisten vorhanden
 - Nur verhältnismäßig wenige Fehler (Fahrer mit 3 Namen bspw.)
- Nutzung der procyclingstat Bibliothek
 - Erzwingt interne HTTP Aufrufe -> sofort geblocked von der Cloudflare Sperre
 - Lösung:
 - Netzwerkhandling an culr_cffi übergeben
 - Modifizierung der Bibliothek zur Laufzeit -> Nutzung nur noch als reiner HTML Parser
- händische Nachbearbeitung der fehlenden Fahrer mit den meisten Rennergebniseinträgen





```
{  
  "birthdate": "1997-3-4",  
  "height": 1.74,  
  "image_url": "images/riders/oh/dq/jhonatan-narvaez-2026.jpg",  
  "name": "Jhonatan Narváez",  
  "nationality": "EC",  
  "place_of_birth": "Playón de San Francisco",  
}
```

Merging



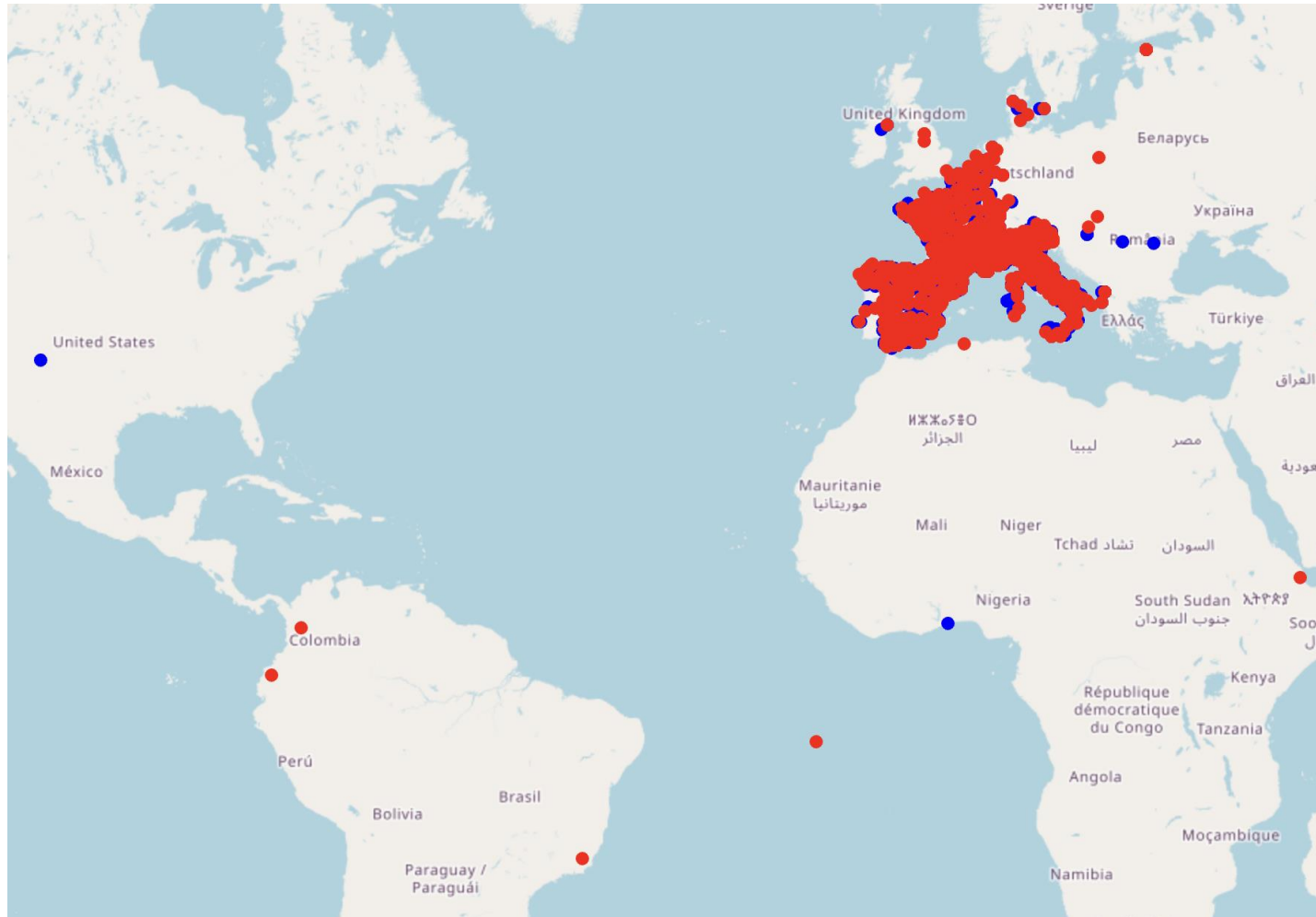
- Problem genesteter Daten
 - Umwandlung in Long Format (Rang, Zeitabstände, Team Historie...)
- Rider_url als eindeutiger Schlüssel -> join (Ergebnisse + Fahrer)
- Ca. 255.000 Zeilen Rohdaten



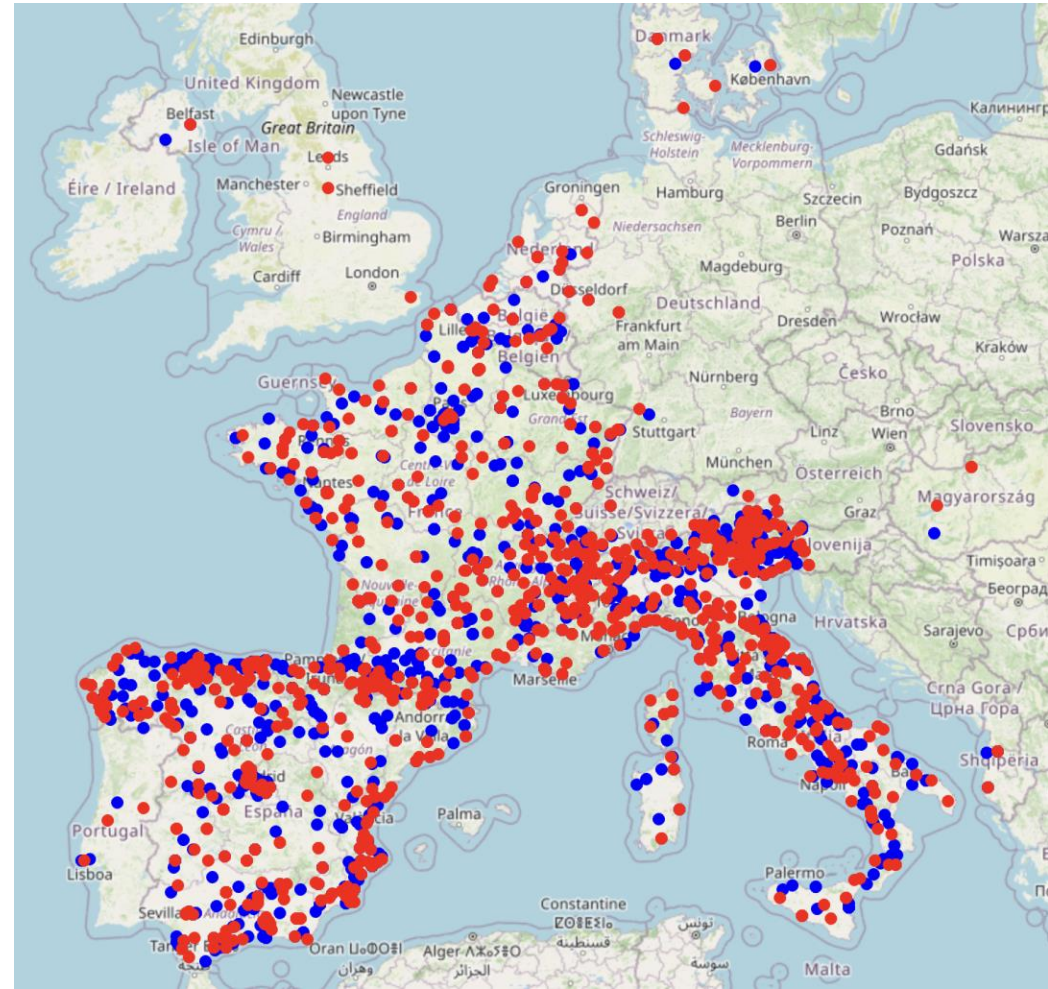
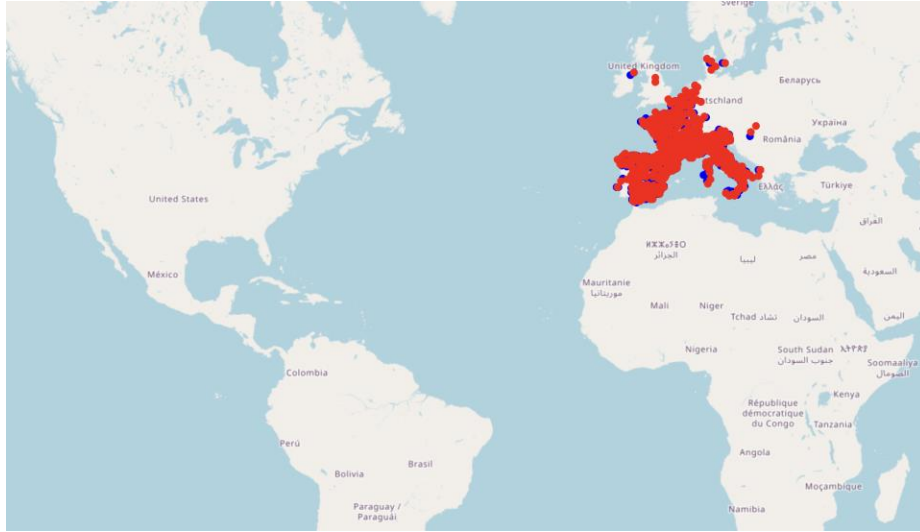
2.2 Geo- und Wetterdaten

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Dresden, 19. Mai 2026

Geodaten



Geodaten



```
ArrivalUndDepartureMitKoordinatendistanz = df1[['departure','arrival', 'departure_lat', 'departure_lon', 'arrival_lat', "arrival_lon", 'distance_km_koordinaten']].drop_duplicates()

display(ArrivalUndDepartureMitKoordinatendistanz)
```

✓ 0.0s

Python

	departure	arrival	departure_lat	departure_lon	arrival_lat	arrival_lon	distance_km_koordinaten
0	Fromentine	Noirmoutier-en-l'Île	46.889671	-2.136662	47.000089	-2.244145	14.741875
189	Challans	Les Essarts	46.847809	-1.877431	48.880997	0.992121	311.302126
378	La Châtaigneraie	Tours	46.650000	-0.739167	47.390047	0.688927	135.979496
567	Tours	Blois	47.390047	0.688927	47.587686	1.333764	53.202643
756	Chambord	Montargis	48.889945	0.609595	47.997863	2.731007	185.259933
945	Troyes	Nancy	48.297163	4.074626	48.693722	6.183410	161.518419
1133	Lunéville	Karlsruhe	48.591616	6.491956	49.006870	8.403420	147.416135
1320	Pforzheim	Gérardmer	48.890934	8.702551	48.070577	6.877696	162.513358
1505	Gérardmer	Mulhouse	48.070577	6.877696	47.746723	7.338994	49.789175
1685	Brignoud	Courchevel - Altiport	45.262528	5.901341	45.397161	6.633462	59.157250

```
ArrivalUndDeparture = df1[['departure','arrival', 'departure_lat', 'departure_lon', 'arrival_lat', "arrival_lon"]].drop_duplicates().head(10)

display(ArrivalUndDeparture)
```

✓ 0.0s

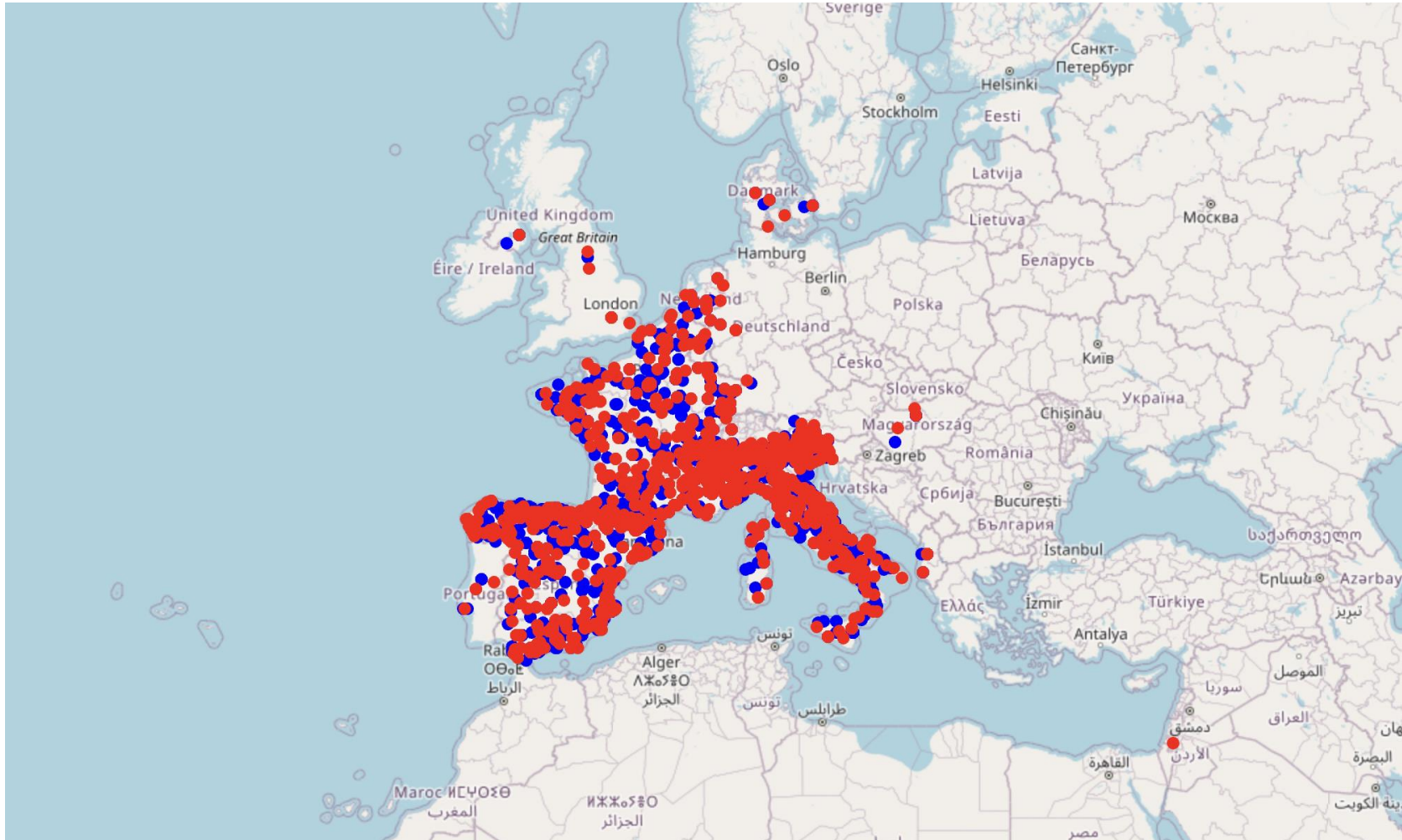
	departure	arrival	departure_lat	departure_lon	arrival_lat	arrival_lon
0	Fromentine	Noirmoutier-en-l'Île	46.889671	-2.136662	47.000089	-2.244145
189	Challans	Les Essarts	46.847809	-1.877431	46.779115	-1.242859
378	La Châtaigneraie	Tours	46.650000	-0.739167	47.390047	0.688927
567	Tours	Blois	47.390047	0.688927	47.587686	1.333764
756	Chambord	Montargis	48.889945	0.609595	47.997863	2.731007
945	Troyes	Nancy	48.297163	4.074626	48.693722	6.183410
1133	Lunéville	Karlsruhe	48.591616	6.491956	49.006870	8.403420
1320	Pforzheim	Gérardmer	48.890934	8.702551	48.070577	6.877696
1505	Gérardmer	Mulhouse	48.070577	6.877696	47.746723	7.338994
1685	Brignoud	Courchevel - Altiport	45.262528	5.901341	45.397161	6.633462

```
print(df1['distance'].max())
```

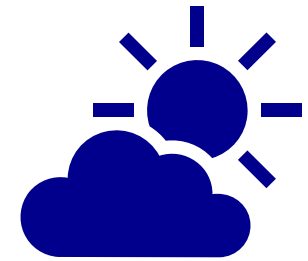
✓ 0.0s

264.0

Geodaten



Wetterdaten



Wetterdaten



Hourly Weather Variables

- | | | |
|---|--|---|
| ☒ Temperature (2 m) | ☐ Weather code | ☒ Wind Speed (10 m) |
| ☒ Relative Humidity (2 m) | ☐ Sea Level Pressure | ☐ Wind Speed (100 m) |
| ☐ Dewpoint (2 m) | ☐ Surface Pressure | ☒ Wind Direction (10 m) |
| ☐ Apparent Temperature | ☐ Cloud Cover Total | ☐ Wind Direction (100 m) |
| ☒ Precipitation (rain + snow) | ☐ Cloud Cover Low | ☐ Wind Gusts (10 m) |
| ☒ Rain | ☐ Cloud Cover Mid | |
| ☐ Snowfall | ☐ Cloud Cover High | |
| ☐ Snow depth | ☐ Reference Evapotranspiration (ET ₀) | |
| | ☐ Vapour Pressure Deficit | |
| ☐ Soil Temperature (0-7 cm) | | |
| ☐ Soil Temperature (7-28 cm) | | |
| ☐ Soil Temperature (28-100 cm) | | |
| ☐ Soil Temperature (100-255 cm) | | |
| ☐ Soil Moisture (0-7 cm) | | |
| ☐ Soil Moisture (7-28 cm) | | |
| ☐ Soil Moisture (28-100 cm) | | |
| ☐ Soil Moisture (100-255 cm) | | |

	departure	departure_temp_mittel	departure_regen_mittel	departure_wind_mittel	departure_luftfeuchte_mittel	departure_niederschlag_mittel	departure_windrichtung_mittel
0	Fromentine	20.0	0.0	12.0	75.25	0.0	214.25
189	Challans	23.82	0.0	12.07	69.0	0.0	328.25
378	La Châtaigneraie	19.52	0.0	21.02	41.75	0.0	278.5
567	Tours	17.53	0.0	15.6	54.25	0.0	220.75
756	Chambord	17.8	0.18	20.88	63.0	0.18	286.25
945	Troyes	16.8	0.12	22.52	67.75	0.12	260.0
1133	Lunéville	16.07	0.25	8.45	79.0	0.25	259.0
1320	Pforzheim	19.35	0.0	6.85	57.75	0.0	51.25
1505	Gérardmer	18.77	0.15	17.3	66.5	0.15	35.5
1685	Brignoud	23.38	0.1	5.5	65.5	0.1	334.0

	arrival	arrival_temp_mittel	arrival_regen_mittel	arrival_wind_mittel	arrival_luftfeuchte_mittel	arrival_niederschlag_mittel	arrival_windrichtung_mittel
0	Noirmoutier-en-l'Île	20.62	0.0	11.55	74.0	0.0	218.25
189	Les Essarts	28.32	0.05	6.22	50.0	0.05	277.25
378	Tours	21.02	0.0	21.8	36.25	0.0	268.0
567	Blois	18.23	0.0	18.17	51.0	0.0	218.5
756	Montargis	20.48	0.03	20.38	52.0	0.03	262.0
945	Nancy	15.15	1.98	17.23	79.5	1.98	240.0
1133	Karlsruhe	17.57	0.25	5.5	68.0	0.25	287.75
1320	Gérardmer	17.32	0.15	9.4	64.5	0.15	30.75
1505	Mulhouse	19.7	0.12	15.57	70.25	0.12	31.75
1685	Courchevel - Altiport	11.93	0.88	3.95	83.75	0.88	301.25



3. Datenexploration

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Univariate Statistiken

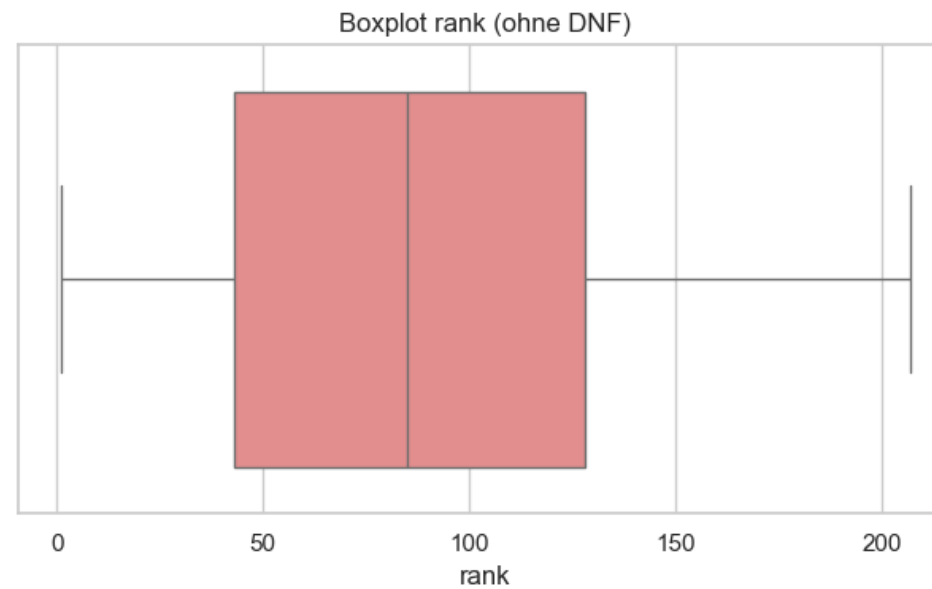
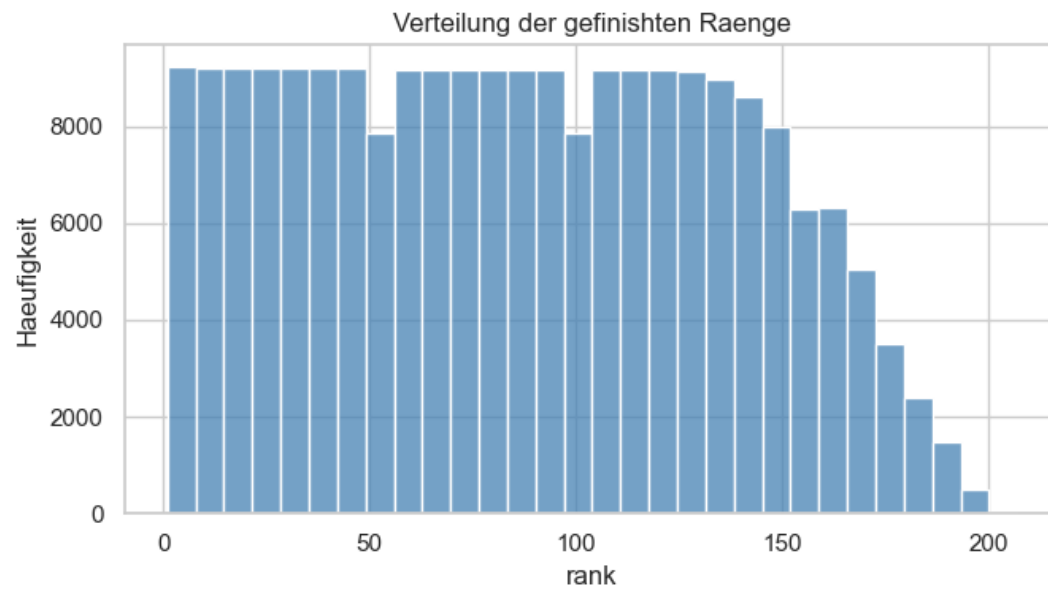
- Aktuell auf Zwischenstand ohne Wetterdaten (--> To-Do)
- Unterscheidung der Spalten nach dtype allein reicht nicht (ordinal vs. Kontinuierlich, ID Spalten, ...)

Einordnung	Beispielspalten	Anzahl Spalten	Angewandte Maße
Numerisch (kontinuierlich)	Distance, weight, avg_speed	14	mean, std, var, min, q1, median, q3, max, iqr, skew, kurtosis, mode, n_unique
Kategorial nominal	Race, nationality, current_team	9	mode, mode_freq, top_5
Datum	Date, birthdate	2	min, max
Ordinal (Integer)	Year, stage_nr	2	Wie numerische Spalten
geo	Departure_lat, departure_lon	4	Min, max, mean
Id_like		4	n_unique, n_duplicates

Zielvariable "rank":

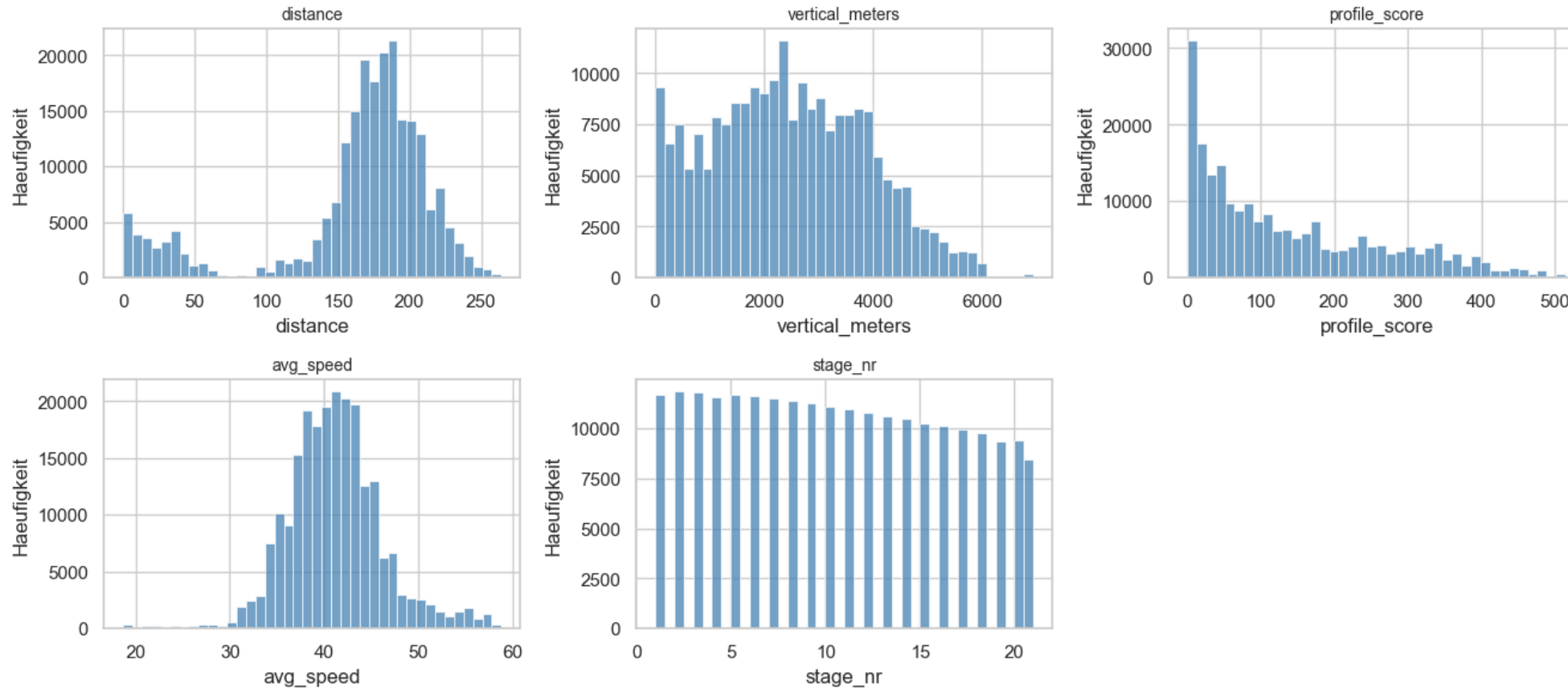
- NaN ist Information: Fahrer, die das Rennen nicht beenden konnten (DNF, DNS, DSQ), bekommen keinen Rang
--> Verzerrung von z.B. mean oder std, da NaN ignoriert werden.
- 1,26 % sind DNF/DNS (rank == NaN)

mean	std	min	max
86,31	50,54	1	207



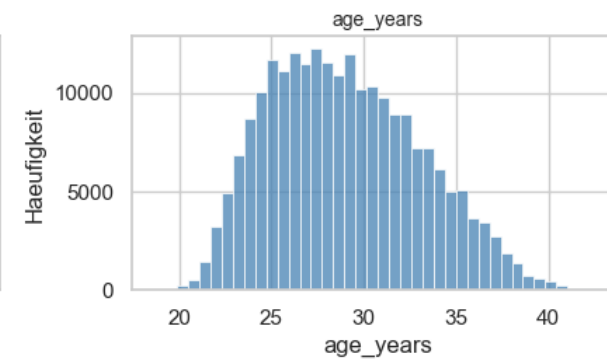
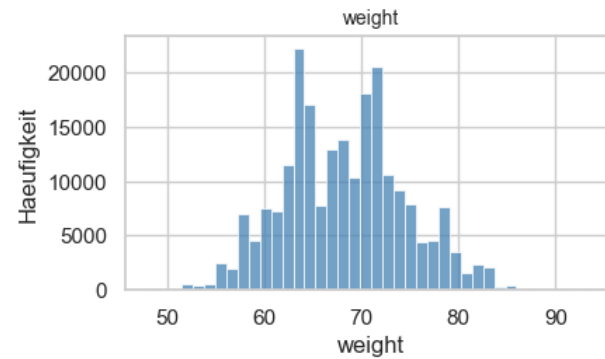
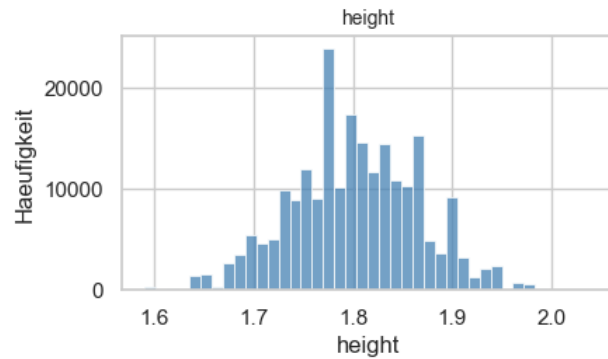
Numerisch: Renn-Features

Renn-Features - Histogramme

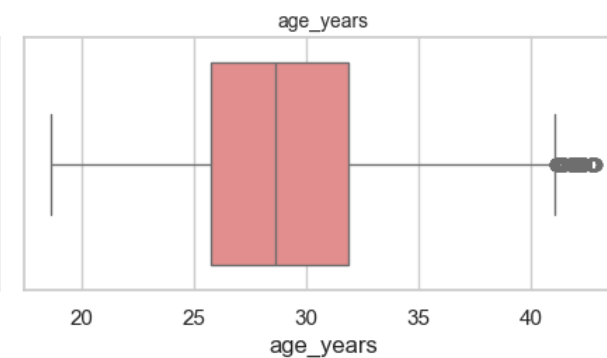
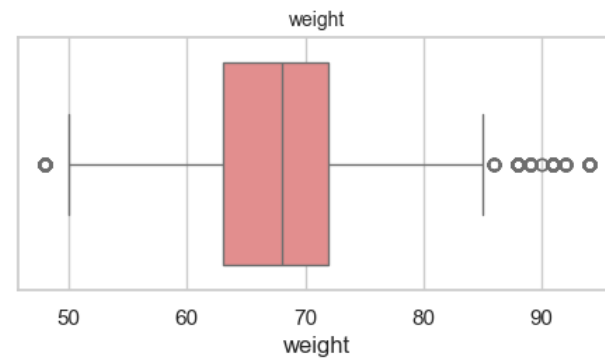
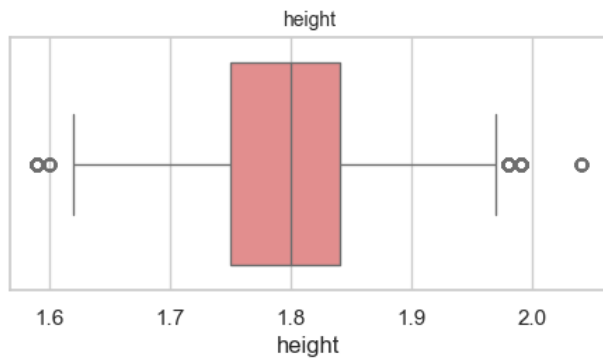


Numerisch: Fahrer-Biometrie

Fahrer-Biometrie - Histogramme

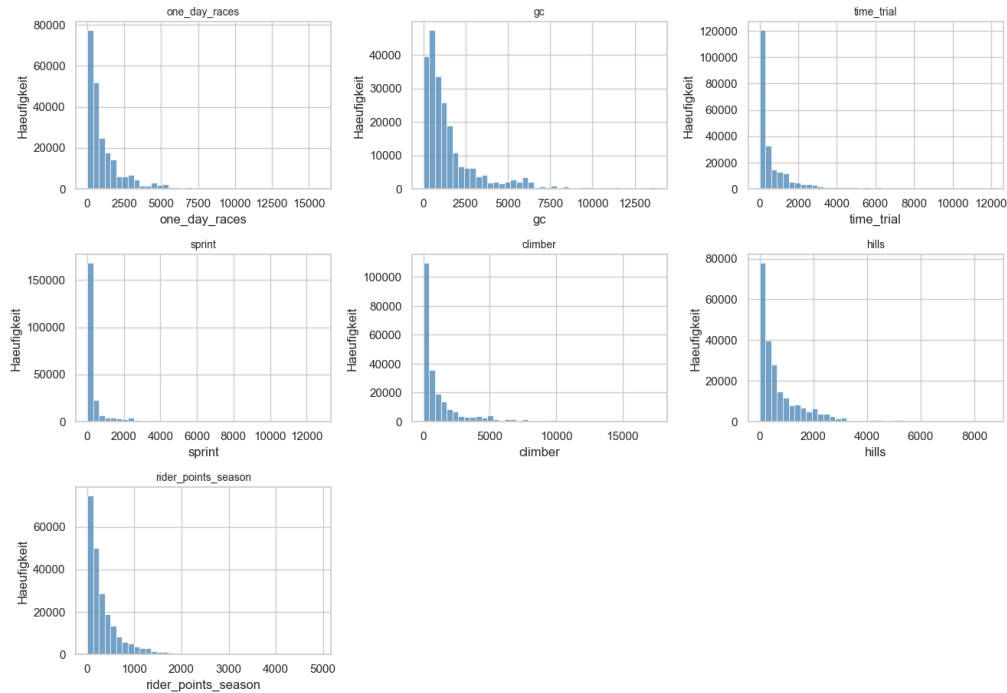


Fahrer-Biometrie - Boxplots

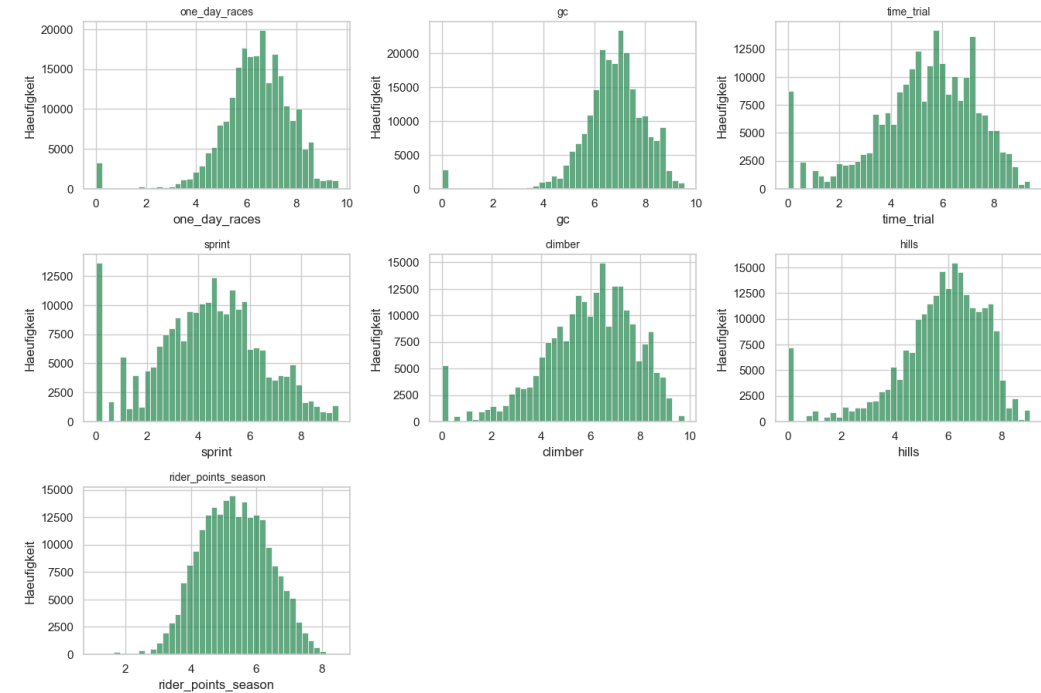


Numerisch: Spezialisierungspunkte (von pcs)

Spezialisierungspunkte - linear



Spezialisierungspunkte - log1p-Skala

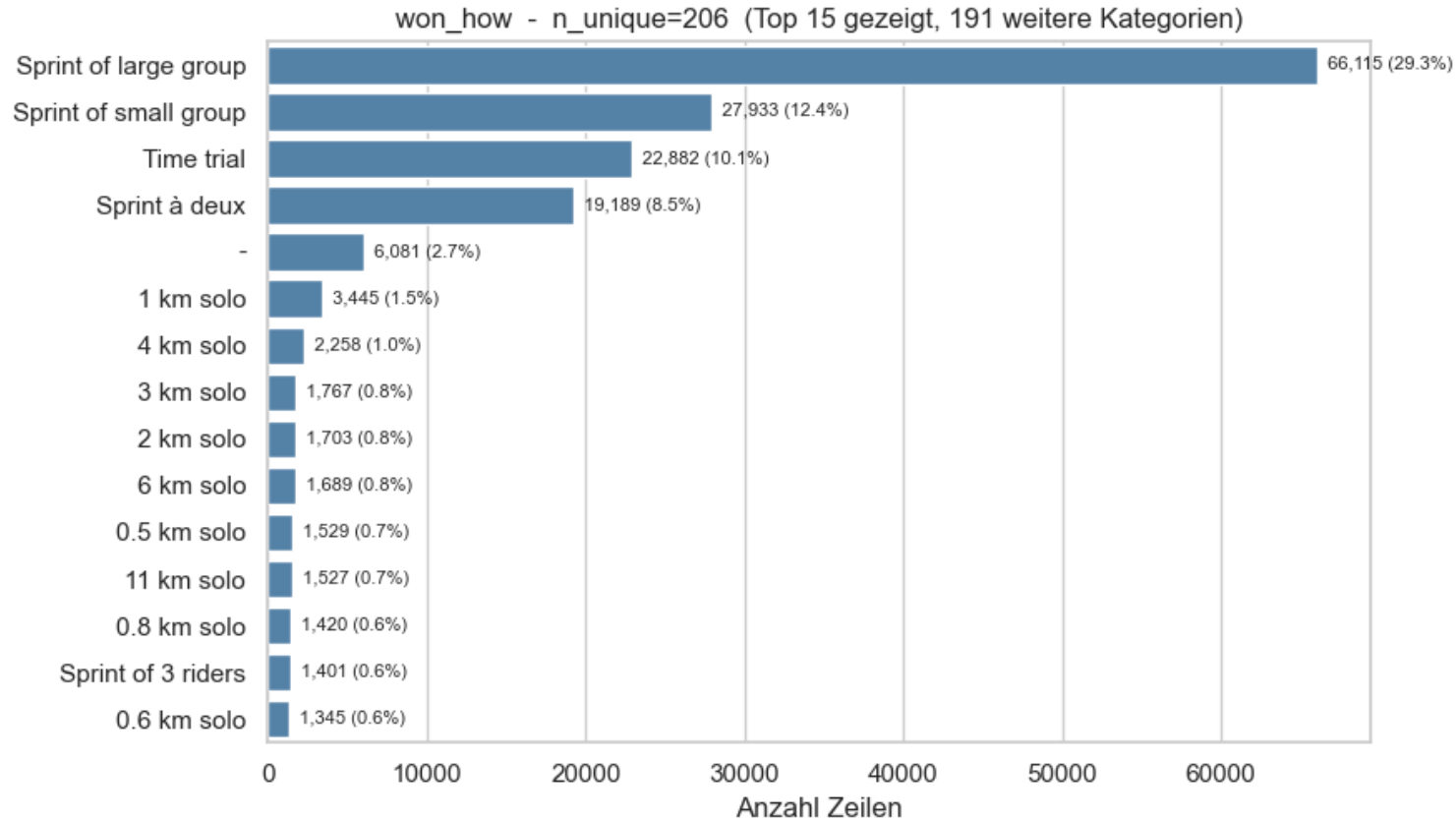


- Stark rechtsschief
- Werte uneindeutig --> in geeignete Skala bringen

Numerisch: Outlier Detection

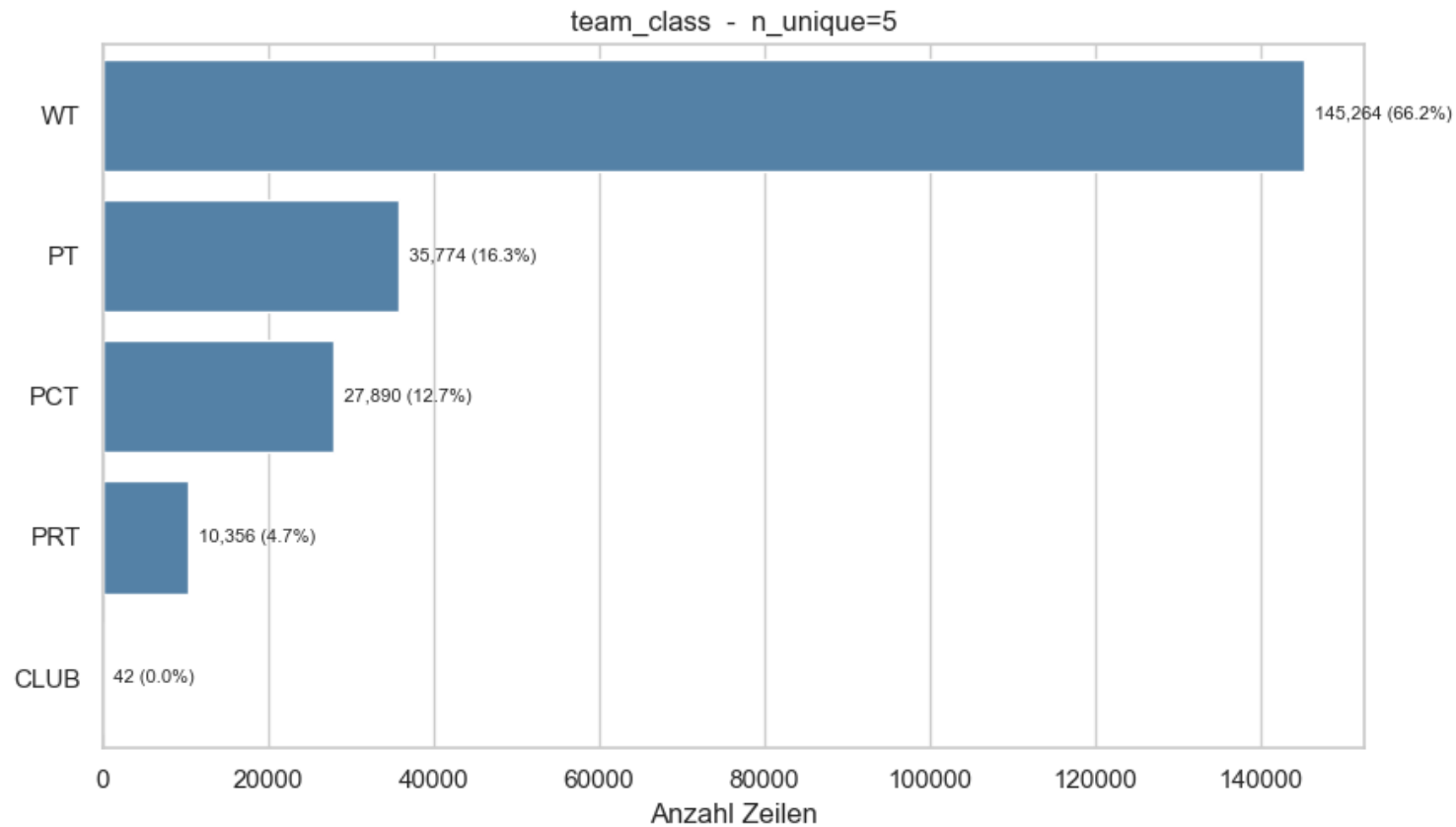
	n	q1	q3	iqr	lower_bound	upper_bound	min	max	n_outliers	outlier_pct
column										
sprint	225692	20.000	322.000	302.000	-433.000	775.000	0.000	12697.00	30388	13.46
distance	225692	153.400	196.500	43.100	88.750	261.150	0.000	264.00	29468	13.06
climber	225692	120.000	1410.000	1290.000	-1815.000	3345.000	0.000	17476.00	25191	11.16
gc	225692	469.000	1838.000	1369.000	-1584.500	3891.500	0.000	13732.00	22096	9.79
time_trial	225692	67.000	923.000	856.000	-1217.000	2207.000	0.000	12022.00	20915	9.27
one_day_races	225692	285.000	1484.000	1199.000	-1513.500	3282.500	0.000	15708.00	18650	8.26
rider_points_season	220930	94.000	444.000	350.000	-431.000	969.000	2.000	4921.00	16800	7.60
hills	225692	135.000	1014.000	879.000	-1183.500	2332.500	0.000	8646.00	15788	7.00
avg_speed	225539	38.066	43.813	5.747	29.446	52.434	18.666	58.83	9125	4.05
rider_rank_season	220930	119.000	567.000	448.000	-553.000	1239.000	1.000	2481.00	2616	1.18
height	221760	1.750	1.840	0.090	1.615	1.975	1.590	2.04	1084	0.49
weight	221760	63.000	72.000	9.000	49.500	85.500	48.000	94.00	768	0.35
vertical_meters	220380	1349.000	3518.000	2169.000	-1904.500	6771.500	0.000	6939.00	173	0.08
profile_score	220380	31.000	229.000	198.000	-266.000	526.000	0.000	515.00	0	0.00

Kategoriale Variablen: won_how



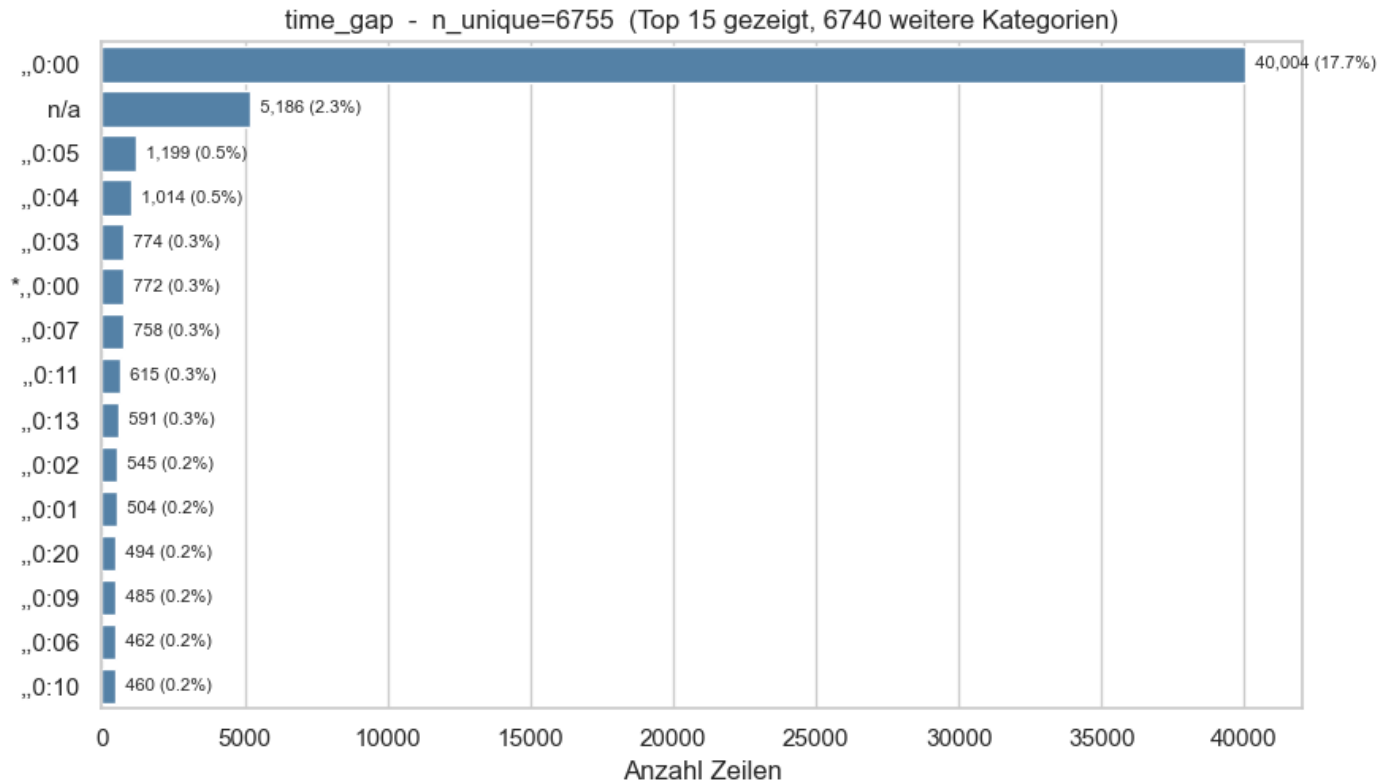
- transformieren

Kategoriale Variablen: team_class



- In ordinale Skala transformieren (World Tour > Pro Tour)

Kategoriale Variablen: time_gap



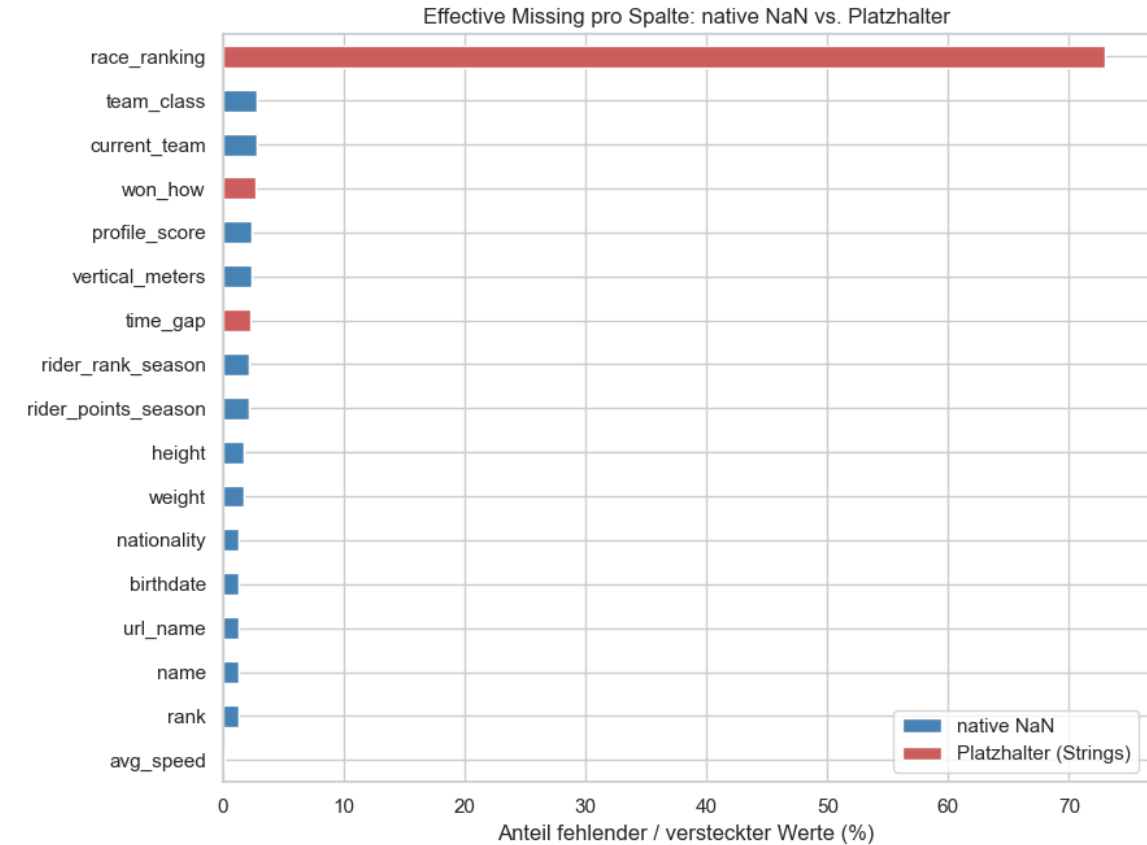
- transformieren

Missing Values

- Datensatz nach weitem Platzhaltern, wie: "-", "none", "unknown", etc. durchsuchen

column	dtype	n_native_nan	n_placeholder_hits	n_effective_missing	effective_missing_pct	placeholders_found
race_ranking	object	0	164674	164674	72.96	'n/a' (164674)
team_class	object	6366	0	6366	2.82	
current_team	object	6366	0	6366	2.82	
won_how	object	0	6081	6081	2.69	'-' (6081)
profile_score	float64	5312	0	5312	2.35	
vertical_meters	float64	5312	0	5312	2.35	
time_gap	object	0	5186	5186	2.30	'n/a' (5186)
rider_rank_season	float64	4762	0	4762	2.11	
rider_points_season	float64	4762	0	4762	2.11	
height	float64	3932	0	3932	1.74	
weight	float64	3932	0	3932	1.74	
nationality	object	2874	21	2895	1.28	'NA' (21)

- race_ranking dropfen



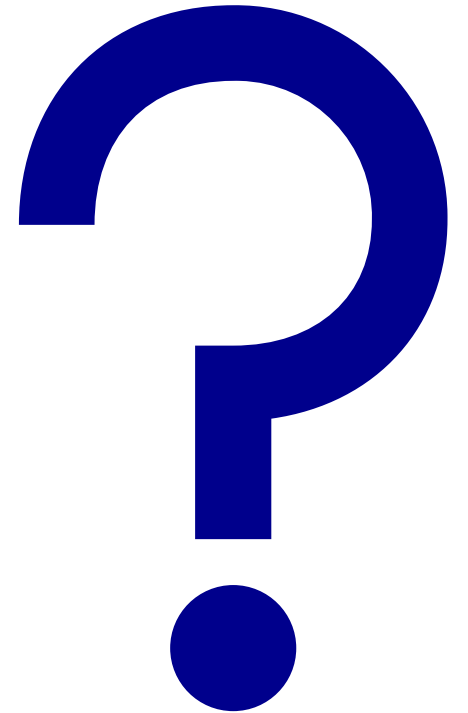


Ausblick

Robert Hendrich, Lukas Pilz und Leon Frey
Dresden, 19. Mai 2026

- Schrittweise Klärung Umgang mit NA's
 - Fehlende Fahrerprofile bereits eliminiert (ca. 2.800 Zeilen)
- Finales Säubern der Daten
- Feature Engineering
- Start Modellanwendung (Feature Selection, HP Tuning etc.)

Gibt es Fragen?



Vielen Dank für Eure
Aufmerksamkeit!

